Computer Organization And Architecture

How and why a computer works

What is a digital computer

A *machine* that can do *work* for people by **carrying out instructions**

1. A sequence of instructions is a *program*
2. Electronic circuits can execute a *limited set of instructions*

These include things like

- adding two numbers
- checking if a number is zero
- copying data from one part of memory to another

These *primitive* instructions, together, form a *language* called **machine language**

So when designing a new computer, one must decide on what instructions to include

Where *less and simpler* instructions mean a *cheaper and faster* computer

It’s difficult to use machine language

so complexity starts to appear

Whenever a it becomes tedious and difficult to use, an **abstraction** is made

Each *layer* of this abstraction has it’s own *complexity*

And this course will be going through most of these layers to more easily understand it

Languages

Fundamentally, what a person wants to *do* and what a computer *can* do have a very large gap

Languages, levels, and Virtual Machines

Let's say that we have *two* languages

- `L1`, a language that's easier for a human
- `L0`, the language that the machine uses

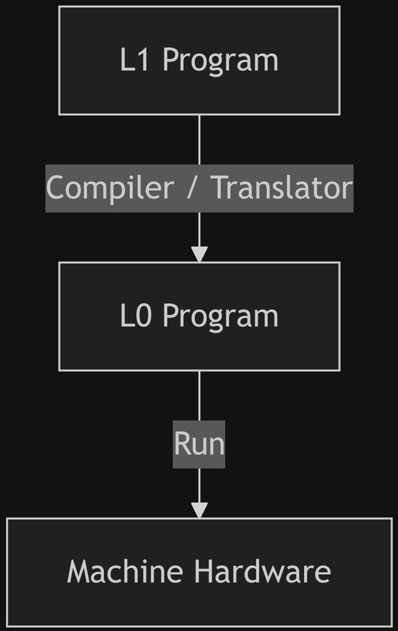
A machine can **only** run programs in the `L0` language

To bridge that gap, there are *two* main methods

Translation

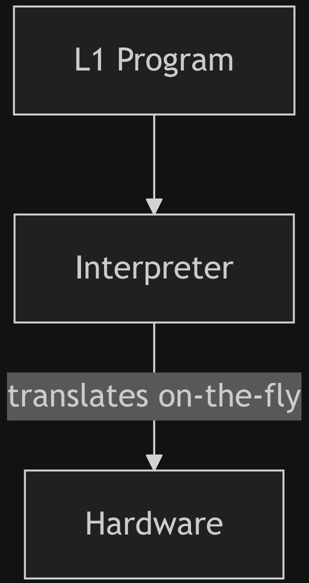
1. Take the `L1` program,
2. take each instruction,
3. replace it with one or more `L0` instructions that are *equivalent*

In the end, you have a fully `L0` program that the machine can run



Interpretation

1. Write an `L0` program that takes in `L1` as input
2. That `L0` program then translates each input as equivalent `L0` code
3. Then that `L0` code is run



Translation vs Interpretation

Virtual Machines

One way of thinking about these different layers is through *virtual machines*

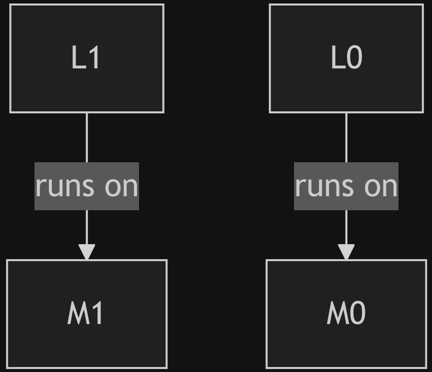
Where each *hypothetical machine* runs a *specific language*

Where `M0` would be a virtual machine that has a **physical** counterpart

And a way to think abouc *translation* or *interpretation* is that we allow `m1` to run on `m0`

So if `L0` is too difficult, we can make `L1`

These languages (`L0` and `L1`) must not be *too* different



Usually this means that `L1` is only *slightly* more convenient than `L0`

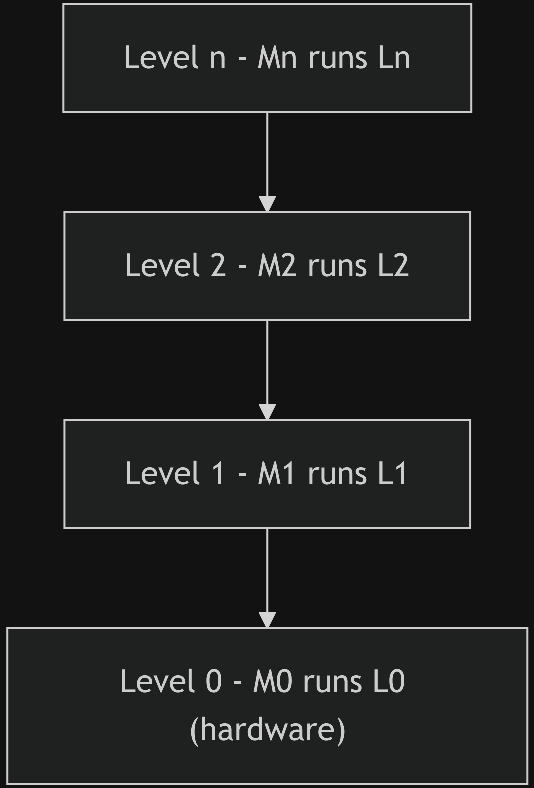
Multilevel machine

A computer with `n` levels can be regarded as `n` different virtual machines

Where programs in `Ln` are

- either *interpreted* by a program running on a *lower machine*, or
- are *translated* to the machine language of a *lower machine*

Note that *level* and *virtual machine* are interchangeable but virtual machine does have a other meanings



Contemporary multi level machines

Contemporary multilevel machines

Most modern computers have two or more levels

We’ll be going through each level

Digital Logic Level

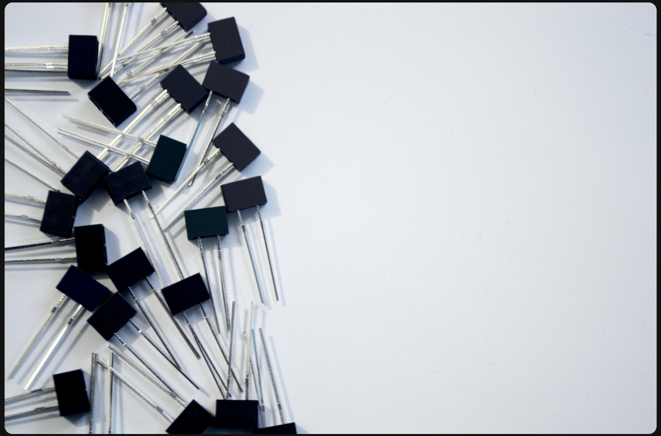
At the *lowest* level, we have

Level 0 - Digital Logic

This has something called **gates** built from analog components like *transistors*

In this level, inputs of **1** and **0**s are present

And combinations of these gates allow for streams of *binary* to hold **memory**, do **mathematics**, and change **state**

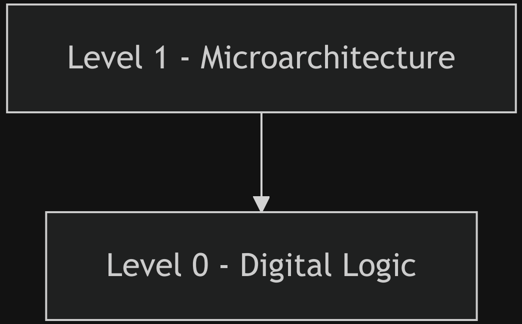


Microarchitecture Level

In this level, we have collections of *registers* and a circuit called the **Arithmetic Logic Unit** (ALU)

The ALU and registers form a *data path*

- *select* one or two registers
- ALU *operates* on them
- *store* result back

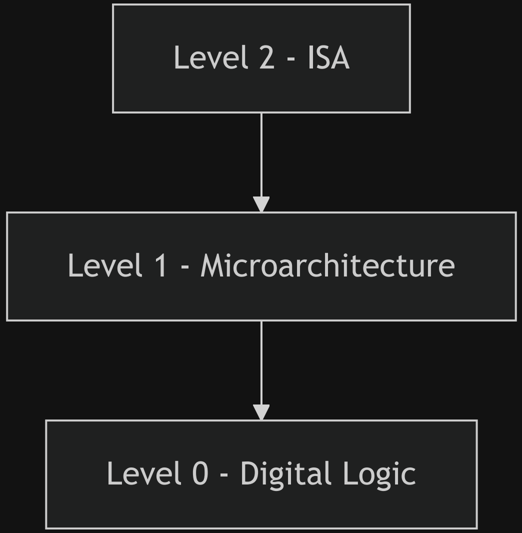


Instruction Set Architecture Level

Every computer manufacturer publishes a manual for each of the computers it sells titled something like "*Machine Language Reference Manual*"

risc-v manual

Level	Example: `ADD R1, R2`
ISA	instruction fetched & decoded
Microarch	registers selected -> ALU adds
Digital Logic	transistors propagate -> result on bus

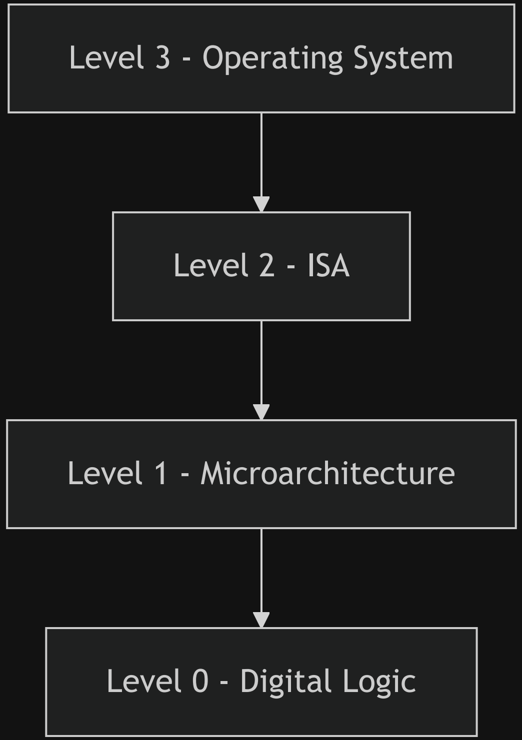


Operating System Level

Adds a set of instructions for

- memory organization
- concurrency and parallelism
- and other features

Note that many operating systems **also** operate *within* the ISA level too



Assembly Language Level

Problem-Oriented Language Level

Summary

- Computers are designed as a **series of levels**, each built on its predecessor
- Each level is a distinct **abstraction** suppressing irrelevant detail to reduce complexity

Architecture vs *Implementation*

- Architecture: visible to the user of that level (data types, operations, features)
- Implementation: how those features are realized (under the hood)

Computer architecture is the study of designing the visible parts of a computer system